

Analysis of the Manifold Hypothesis and Latent Space Capacity in Variational Autoencoders

Chika Obata[†], Kenya Jin'no[†]

[†]Department of Intelligent Systems, Faculty of Information Technology, Tokyo City University
1-28-1, Tamazutsumi, Setagaya-ku, Tokyo 158-8557, Japan,

Abstract—The manifold hypothesis, which underpins the success of deep learning, assumes that high-dimensional data are distributed on low-dimensional manifolds; yet it is still unclear to what extent deep generative models can learn this structure quantitatively. In this study we experimentally verify how well a Variational Autoencoder (VAE) projects the manifold structure of input data into its latent space by analysing the *intrinsic dimension* (ID). Multiple VAE models with different architectures (fully-connected vs. convolutional) and bottleneck dimensions D were trained on MNIST, and the ID of their latent spaces was estimated using principal component analysis (PCA) and the nonlinear TwoNN estimator. For every setting, the latent ID converged to a value *smaller* than the input ID and was insensitive to D or the architecture. These results quantitatively confirm that VAEs not only preserve but further compress the intrinsic manifold structure of data, providing practical guidance for selecting latent-space capacity in VAE design.

1. Introduction

Recent breakthroughs in deep learning hinge on an ability to learn meaningful low-dimensional representations from high-dimensional data. The *manifold hypothesis*—that real-world data such as images and text lie on low-dimensional manifolds embedded in the ambient space—is widely accepted [1, 2]. A Variational Autoencoder (VAE) is a probabilistic latent-variable model ... during decoding [3]. Because the Kullback–Leibler (KL) regulariser ... keeps the latent space continuous and smooth [4], a VAE is well suited to manifold learning. Nevertheless, many open questions remain, including the *disentanglement* of latent factors [5] and *manifold overfitting* [6]. We quantitatively examine how faithfully a VAE's latent space reflects the manifold structure of input data by estimating the intrinsic dimension (ID). Specifically, we focus on:

1. Whether the latent ID converges to the input ID;

2. How the relation between the bottleneck dimension D and the input ID M ($D > M$, $D \approx M$, $D < M$) influences manifold learning;
3. Whether architectural differences (fully-connected vs. convolutional) affect the stability of the learned manifold.

2. Background and Related Works

2.1. Manifold Hypothesis and Intrinsic Dimension

The manifold hypothesis states that an observed data point $\mathbf{x} \in \mathbb{R}^N$ is distributed on a manifold $\mathcal{M}$ of much lower dimension M [1, 2]. Estimating M —the intrinsic dimension—has been tackled by global linear methods such as PCA [7] (prone to overestimation for nonlinear manifolds) and local nonlinear methods such as TwoNN [8], which exploits the distribution of first- and second-nearest neighbour distances and is robust to noise. Recent work measures IDs inside deep networks [Ansuini2019] and evaluates the manifold quality of generative models [12].

2.2. VAE and Geometry of the Latent Space

A VAE is trained to maximise the ELBO [3]; the KL term pushes the latent space $\mathbf{z}$ towards a normal prior, enforcing continuity and smoothness [4]. However, *posterior collapse* [14] and manifold overfitting [6] can arise depending on the KL weight and decoder capacity. Variants such as β -VAE [4, 5] modulate the KL term to promote disentanglement. Analyses based on information imbalance or intrinsic dimension [13], phase-transition phenomena [17], and ID dynamics during training [18] have deepened our understanding of the latent geometry.

3. Experiments

We resized the MNIST handwritten-digit images to 32×32 pixels and trained class-specific VAEs. The input dimension is $N = 1024$. Six architectures were considered: four fully-connected models (model1–model4) and two convolutional models (model5, model6). For each architecture we performed 10 classes $\times$ 4 random seeds, totalling $6 \times 10 \times 4 = 240$ trained VAEs.

ORCID iDs First Author:  0009-0001-1209-0918, Second Author:  0000-0002-0431-5769, :



This work is licensed under a Creative Commons Attribution Non Commercial, No Derivatives 4.0 License. ©IEICE 2025

Table 1: Intrinsic dimension of each MNIST class (TwoNN vs. PCA).

Label	TwoNN	PCA (Broken-Stick)
0	15.30	35
1	13.01	22
2	15.29	46
3	16.40	46
4	14.60	42
5	15.89	44
6	14.20	39
7	13.26	37
8	16.68	50
9	14.15	39

Bottleneck dimensions. Three bottleneck settings were examined:

- D_0 : the *self-converged* latent ID discovered during preliminary runs;
- D_1 : the input-space ID estimated by TwoNN/PCA;
- D_2 : an enlarged value defined as $(D_1 + N)/2$.

3.1. Latent-Space Intrinsic Dimension

Across all conditions, the latent ID converged to values smaller than the input ID ($M \approx 13$ –17) regardless of D or architecture. Convolutional models exhibited IDs 0.5–2 dimensions higher than fully-connected models, but the difference was within statistical error.

3.2. Reconstruction Error

Table 7 lists the mean-squared reconstruction error (MSE). Architectural effects were dominant: convolutional models achieved 30–50% lower MSE than fully-connected ones. Overly large bottleneck sizes (D_2) *increased* MSE, suggesting that excessive capacity leads to overfitting noise or imbalance with the KL regulariser.

4. Discussion

Our main finding is that a VAE autonomously forms a latent manifold of *lower* dimensionality than the input manifold. This modifies the implicit assumption of *ID preservation* and suggests that VAEs learn even more compact representations by discarding redundancy and noise irrelevant to reconstruction. When the bottleneck size exceeds the input ID, reconstruction error increases, implying that excessive capacity encourages the model to memorise noise or destabilises the KL balance. Convolutional architectures require slightly higher latent IDs—likely because local convolutions encode finer spatial structure—but the core mechanism of dimensionality reduction appears architecture-agnostic. Hence, the input ID is a useful first-choice for setting D , and tuning around $D \approx M$ yields stable training and good generalisation. Future work should

examine the effect of KL weighting in β -VAEs, track ID dynamics during training, and test the generality of the phenomenon on datasets such as CIFAR-10 and ImageNet.

5. Conclusion

By analysing 240 VAEs we systematically investigated how the intrinsic dimension of latent spaces is determined.

1. Latent IDs converge to values *smaller* than the input IDs, indicating efficient dimensionality reduction.
2. Convolutional models exhibit marginally higher IDs than fully-connected ones, but the gap is limited.
3. Bottleneck sizes near the input ID achieve the lowest reconstruction error.

These insights recommend choosing $D \approx M$ while carefully balancing compression and generalisation. Estimating the input ID in advance can thus guide the construction of more efficient and robust VAEs.

Acknowledgements

This work was supported by JSPS KAKENHI Grant Numbers JP23K11266, JP23K28077, and JP24K15115, and by the Collaborative Research Project [R06/B14] “Research on Representation Learning in Deep Learning” at the Research Institute of Electrical Communication, Tohoku University.

References

- [1] J.B. Tenenbaum, V. de Silva, and J.C. Langford, “A global geometric framework for nonlinear dimensionality reduction,” *Science*, vol. 290, no. 5500, pp. 2319–2323, 2000.
- [2] S. Fefferman, S. Mitter, and H. Narayan, “Testing the manifold hypothesis,” *J. Amer. Math. Soc.*, vol. 29, no. 4, pp. 983–1049, 2016.
- [3] D. P. Kingma and M. Welling, “Auto-Encoding Variational Bayes,” arXiv:1312.6114, 2013.
- [4] C. P. Burgess *et al.*, “Understanding disentangling in β -VAE,” arXiv:1804.03599, 2018.
- [5] R. T. Q. Chen, X. Li, R. Grosse, and D. Duvenaud, “Isolating sources of disentanglement in variational autoencoders,” in *Advances in Neural Information Processing Systems 31*, 2018.
- [6] B. Dai and D. Wipf, “Diagnosing and enhancing VAE models,” in *Proc. ICLR*, 2019.
- [7] I. T. Jolliffe, *Principal Component Analysis*, 2nd ed., Springer, 2002.

Table 2: Latent IDs for D_1 (TwoNN) condition.

Label	FC				CNN	
	model1	model2	model3	model4	model5	model6
0	6.63 ± 0.25	5.85 ± 0.22	7.37 ± 0.09	6.96 ± 0.13	8.90 ± 0.18	7.95 ± 0.13
1	4.02 ± 0.30	4.02 ± 0.15	5.07 ± 0.08	5.74 ± 0.03	6.16 ± 0.06	6.24 ± 0.19
2	7.73 ± 0.12	7.33 ± 0.12	8.34 ± 0.14	7.94 ± 0.18	8.76 ± 0.13	8.24 ± 0.09
3	8.04 ± 0.18	7.10 ± 0.42	8.73 ± 0.25	8.08 ± 0.21	9.50 ± 0.28	9.58 ± 0.29
4	6.74 ± 0.10	6.77 ± 0.09	7.43 ± 0.10	7.24 ± 0.24	8.99 ± 0.28	8.12 ± 0.11
5	7.63 ± 0.12	6.78 ± 0.10	8.31 ± 0.06	7.65 ± 0.23	9.41 ± 0.12	9.10 ± 0.23
6	6.91 ± 0.16	6.78 ± 0.19	7.39 ± 0.13	7.33 ± 0.12	8.86 ± 0.13	8.40 ± 0.03
7	5.95 ± 0.14	6.00 ± 0.03	6.76 ± 0.11	6.84 ± 0.05	7.66 ± 0.18	7.19 ± 0.14
8	8.29 ± 0.21	7.20 ± 0.37	8.68 ± 0.14	8.67 ± 0.20	9.24 ± 0.27	8.93 ± 0.10
9	7.05 ± 0.16	6.47 ± 0.22	7.81 ± 0.04	7.51 ± 0.09	8.95 ± 0.27	8.56 ± 0.21

Table 3: Latent IDs for D_2 (TwoNN) condition.

Label	FC				CNN	
	model1	model2	model3	model4	model5	model6
0	5.84 ± 0.06	5.44 ± 0.22	6.35 ± 0.03	8.02 ± 0.02	9.71 ± 0.29	7.88 ± 0.10
1	3.92 ± 0.02	4.11 ± 0.06	4.81 ± 0.04	8.07 ± 0.10	6.56 ± 0.14	5.77 ± 0.10
2	6.72 ± 0.16	6.18 ± 0.17	7.63 ± 0.08	9.49 ± 0.03	10.28 ± 0.13	8.89 ± 0.04
3	6.92 ± 0.08	5.33 ± 0.15	7.55 ± 0.15	9.87 ± 0.03	10.56 ± 0.15	9.90 ± 0.11
4	6.24 ± 0.05	5.60 ± 0.05	6.76 ± 0.05	8.32 ± 0.07	9.53 ± 0.15	8.04 ± 0.13
5	6.26 ± 0.06	5.46 ± 0.13	7.12 ± 0.07	9.90 ± 0.05	10.76 ± 0.18	9.56 ± 0.12
6	6.32 ± 0.15	5.50 ± 0.14	7.13 ± 0.18	8.68 ± 0.04	9.04 ± 0.17	8.24 ± 0.10
7	5.55 ± 0.05	4.64 ± 0.05	6.31 ± 0.15	7.81 ± 0.03	8.46 ± 0.06	7.15 ± 0.09
8	6.51 ± 0.10	5.15 ± 0.19	8.17 ± 0.19	10.43 ± 0.09	10.65 ± 0.11	9.53 ± 0.16
9	6.21 ± 0.15	5.16 ± 0.16	6.78 ± 0.07	8.54 ± 0.02	9.11 ± 0.16	7.98 ± 0.13

- [8] E. Facco, M. d’Errico, A. Rodriguez, and A. Laio, “Estimating the intrinsic dimension of datasets by a minimal neighbourhood information,” *Sci. Rep.*, vol. 7, 12140, 2017.
- [9] E. Levina and P. J. Bickel, “Maximum likelihood estimation of intrinsic dimension,” in *Advances in Neural Information Processing Systems 17*, 2005.
- [10] A. Ansuini, A. Laio, J. I. Mejía-Lastra, and A. Rodriguez, “Intrinsic dimension of data representations in deep neural networks,” in *Proc. NeurIPS*, 2019.
- [11] P. E. Pope *et al.*, “The intrinsic dimension of images and its impact on learning,” in *Proc. ICLR*, 2021.
- [12] G. Loaiza-Ganem *et al.*, “Diagnosing and fixing manifold overfitting in deep generative models,” *Trans. Mach. Learn. Res.*, 2022.
- [13] C. Camboulin, D. Doimo, and A. Glielmo, “Understanding Variational Autoencoders with Intrinsic Dimension and Information Imbalance,” arXiv:2411.01978, 2024.
- [14] K. Xu *et al.*, “Understanding posterior collapse in generative latent-variable models,” in *Proc. ICLR*, 2019.
- [15] Y. LeCun, L. Bottou, Y. Bengio, and P. Haffner, “Gradient-based learning applied to document recognition,” *Proc. IEEE*, vol. 86, no. 11, pp. 2278–2324, 1998.
- [16] L. Deng, “The MNIST database of handwritten digit images for machine learning research,” *IEEE Signal Processing Mag.*, vol. 29, no. 6, pp. 141–142, 2012.
- [17] A. Lucas *et al.*, “Analysing phase transitions in variational autoencoders,” arXiv:2106.05464, 2021.
- [18] C. Zhao, X. Li, K. Narasimhan, and D. Duvenaud, “On learning intrinsic dimension of data representations,” arXiv:2203.05796, 2022.

Table 4: Latent IDs for D_1 (PCA) condition.

Label	FC				CNN	
	model1	model2	model3	model4	model5	model6
0	7.75 ± 0.50	7.00 ± 0.00	11.25 ± 0.50	11.50 ± 0.58	13.75 ± 1.26	15.00 ± 1.83
1	4.00 ± 0.00	4.00 ± 0.00	4.50 ± 0.58	5.00 ± 0.00	6.25 ± 0.96	6.50 ± 0.58
2	10.50 ± 0.58	9.50 ± 0.58	13.25 ± 0.50	14.00 ± 0.00	21.00 ± 0.82	22.25 ± 0.96
3	10.00 ± 0.00	8.25 ± 0.50	12.00 ± 0.00	11.00 ± 0.00	18.50 ± 1.29	19.00 ± 1.41
4	9.00 ± 0.00	8.00 ± 0.00	9.75 ± 0.50	11.00 ± 0.00	14.75 ± 0.50	15.25 ± 0.50
5	9.25 ± 0.50	7.75 ± 0.50	11.50 ± 0.58	9.25 ± 0.50	19.50 ± 0.58	19.25 ± 0.50
6	8.25 ± 0.50	6.75 ± 0.50	9.75 ± 0.50	10.00 ± 0.00	14.00 ± 0.00	14.75 ± 0.96
7	7.00 ± 0.00	7.00 ± 0.00	8.50 ± 0.58	8.00 ± 0.00	14.00 ± 1.41	12.00 ± 0.82
8	9.75 ± 0.50	6.75 ± 0.50	12.50 ± 0.58	13.00 ± 0.00	18.25 ± 0.96	17.00 ± 0.00
9	8.50 ± 0.58	7.25 ± 0.50	9.00 ± 0.00	9.00 ± 0.00	13.00 ± 0.00	13.25 ± 0.96

Table 5: Latent IDs for D_2 (PCA) condition.

Label	FC				CNN	
	model1	model2	model3	model4	model5	model6
0	9.75 ± 0.50	7.00 ± 0.00	12.00 ± 0.00	18.00 ± 0.00	21.00 ± 1.16	16.00 ± 0.00
1	5.00 ± 0.00	4.50 ± 0.58	7.00 ± 0.00	8.00 ± 0.00	13.25 ± 0.96	8.25 ± 0.50
2	11.00 ± 0.82	7.00 ± 0.00	15.00 ± 0.00	21.25 ± 0.50	28.50 ± 0.58	21.00 ± 0.00
3	10.25 ± 0.50	6.50 ± 0.58	13.00 ± 0.00	19.00 ± 0.00	27.25 ± 0.50	19.25 ± 0.50
4	10.00 ± 0.00	7.00 ± 0.82	12.25 ± 0.50	16.25 ± 0.50	24.75 ± 1.26	16.25 ± 0.50
5	9.25 ± 0.50	6.50 ± 0.58	12.00 ± 0.00	19.00 ± 0.00	26.50 ± 0.58	18.75 ± 0.96
6	10.00 ± 0.00	6.50 ± 0.58	13.25 ± 0.50	17.75 ± 0.50	21.00 ± 0.00	15.50 ± 0.58
7	9.25 ± 0.50	7.00 ± 0.00	12.00 ± 0.00	15.00 ± 0.00	21.50 ± 1.29	14.50 ± 0.58
8	10.00 ± 0.00	5.50 ± 0.58	12.75 ± 0.50	20.00 ± 0.00	27.00 ± 1.16	17.50 ± 0.58
9	9.00 ± 0.00	6.50 ± 0.58	12.00 ± 0.00	16.75 ± 0.50	19.75 ± 0.50	14.50 ± 1.00

Table 6: Latent IDs for D_0 (self-converged) condition.

Label	FC				CNN	
	model1	model2	model3	model4	model5	model6
0	6.04 ± 0.06	5.63 ± 0.11	5.71 ± 0.08	5.87 ± 0.21	5.89 ± 0.09	5.77 ± 0.05
1	3.54 ± 0.51	3.74 ± 0.11	4.13 ± 0.04	4.45 ± 0.08	4.65 ± 0.02	4.20 ± 0.21
2	6.73 ± 0.07	6.74 ± 0.16	6.64 ± 0.05	6.53 ± 0.08	6.59 ± 0.06	6.38 ± 0.13
3	7.45 ± 0.04	6.73 ± 0.22	6.89 ± 0.19	6.64 ± 0.34	7.33 ± 0.16	7.09 ± 0.19
4	5.68 ± 0.09	5.87 ± 0.06	5.84 ± 0.09	5.99 ± 0.13	6.05 ± 0.05	5.73 ± 0.11
5	6.91 ± 0.17	6.42 ± 0.07	6.44 ± 0.11	6.19 ± 0.38	7.00 ± 0.06	6.96 ± 0.16
6	6.30 ± 0.21	5.97 ± 0.30	5.90 ± 0.15	5.89 ± 0.11	6.28 ± 0.10	5.89 ± 0.12
7	5.16 ± 0.07	5.34 ± 0.30	5.23 ± 0.07	5.01 ± 0.12	5.32 ± 0.05	5.40 ± 0.04
8	7.57 ± 0.16	6.64 ± 0.29	7.08 ± 0.19	7.29 ± 0.06	7.81 ± 0.19	7.34 ± 0.28
9	6.19 ± 0.44	6.01 ± 0.07	6.08 ± 0.05	5.98 ± 0.24	6.50 ± 0.14	5.94 ± 0.08

Table 7: MSE for each model and bottleneck size.

Model	D_0	D_1	D_2
model1	0.0140	0.0149	0.0239
model2	0.0165	0.0190	0.0271
model3	0.0157	0.0162	0.0236
model4	0.0282	0.0234	0.0262
model5	0.0106	0.0073	0.0097
model6	0.0134	0.0096	0.0165